

Original Research

Mitigating hallucinations in synthesized clinical texts to improve multimodal deep learning for dermatology

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ABSTRACT

Objective: Despite recent advancements in the development of foundation models and multimodal (MM) architectures in dermatology, their translation to clinical practice remains limited by the scarcity of large-scale multimodal (MM) datasets, as most publicly available resources are small, unimodal, and lack expressive clinical text. This paper investigates strategies to synthesize and exploit clinical notes paired with dermatological images to effectively train a MM architecture, focusing on solutions to limit the inherently hallucinated contents introduced by Large Language Models (LLMs) and to identify conditions under which synthetic clinical notes can be reliably leveraged.

Methods: The paper proposes a MM architecture trained on real dermatological images paired with LLM-synthesized clinical notes. We systematically evaluate different note generation strategies, including metadata-guided prompting, alignment of image representations with specific keywords, sentence-level filtering of clinical notes, network architectural designs. Experiments involve 16,000 image-note couples collected from six public datasets for model training and over 37,000 images from fifteen public datasets as external data for generalization assessment. Performance is assessed on cross-modal retrieval and zero-shot learning tasks to quantify robustness and generalization.

Results: Results show that metadata inclusion into the prompts reduces the hallucinations within LLM outputs, providing more reliable notes. The resulting MM model trained with these notes show superior performance on multiple downstream tasks.

Conclusion: Synthesized clinical notes can be paired with real dermatology images under specific conditions, providing a valuable resource to develop foundation models that can help reduce the dermatologists' workload.

1. Introduction

Dermatology is a medical domain that has recently shown rapidly growing interest in the application of Artificial Intelligence (AI), particularly foundation models and Large Language Models (LLMs), to support dermatologists in the analysis and diagnosis of skin lesions; however, significant challenges still limit the development of reliable models suitable for routine clinical practice [1–5].

Skin diseases represent a major global health burden, affecting a large fraction of the worldwide population and ranking among one of the leading causes of non-fatal disease across different regions and age groups [6–11]. Although most skin lesions are benign, a clinically relevant subset includes malignant lesions, that require accurate and often time-consuming evaluation [11,12]. Clinical assessment remains challenging even for experienced dermatologists due to the lesion

appearance heterogeneity (i.e. color, border appearance, and morphological structures), spanning between benign nevi and malignant entities such as melanomas [13–17]. Early diagnosis is hindered by the limited availability of dermatology specialists, limiting their capability to cover the increasing demand for care [9,18–20]. Consequently, patients often face delayed diagnosis or are evaluated by non-specialist clinicians, increasing the risk of missed or late-detected malignancies [16,21–24]. These constraints have led to increasing interest in automatic approaches to support clinical decision-making in dermatology, with an increased interest in AI-based methods [4,5,25–27].

Recent advances in AI, especially Deep Learning (DL) methods, have demonstrated strong potential, with models achieving dermatologist-level performance in tasks such as lesion classification and risk assessment [11,28,28–30]. DL methods are particularly effective when integrating multiple medical data modalities, for example combining images

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with free-text clinical notes and structured patient information [31,32]. Multimodal (MM) learning aims to model richer medical representations, leveraging the complementary nature of different data types, such as the visual and textual modalities identified above, particularly because they may include meaningful and often non-redundant clinical information [31–33]. Language integration enables text-based querying of image datasets [34–36], facilitating application transfer across tasks and reducing the development of algorithms based on rigid categorical labels [33,37]. These properties have enabled the increasing success of large-scale foundation models pre-trained on vast multimodal corpora to be extended to the development of dermatology-specific multimodal models [1,23,32,34,38–43].

However, the performance of dermatology-specific multimodal models is limited by data availability and characteristics [26,44]. Most publicly available dermatology datasets are limited in size and unimodal, primarily including images paired with categorical labels or minimal heterogeneous metadata, while expressive textual descriptions are often absent [27,44]. This limitation is a consequence of practical constraints: free-text clinical notes may be difficult to share due to privacy legislations, are highly heterogeneous in style and content, and are related to specific clinical workflow documentation practices [45,46]. As a result, assembling large-scale, high-quality image-text pairs through manual curation is often infeasible, preventing multimodal learning from scaling through traditional data collection [44,45].

Existing multimodal dermatology datasets that include images paired with textual descriptions introduce further limitations [23,26]. Large-scale collections often contain heterogeneous and noisy images, including photographs or web-scraped material in which lesions appear only as background elements, which reduces the algorithm suitability for use in controlled clinical settings, such as dermoscopy [47,48]. In addition, associated textual descriptions are typically generic and non-expressive, often limited to lesion type or diagnosis without detailed characterization of visual attributes [23,48]. Consequently, the text modality provides limited contribution, restricting the effective use of multimodal architectures and limiting the transfer of foundation model advances to dermatology [26,32,47].

The synthesis of clinical notes from images using LLMs has emerged as a practical necessity to enable multimodal learning under realistic data constraints [1,45,49]. Synthetic clinical notes provide a scalable mechanism to introduce textual content into large image-only dermatology datasets, potentially enabling MM learning and more flexible training paradigms [33,44,49]. However, LLM-generated text is inherently affected by hallucinations and noisy content, introducing information that may be inaccurate or unsupported by the visual input [50–53]. These hallucinations are often systematic, reflecting strong information learned during pre-training, and can hinder multimodal alignment if used naively, leading to spurious correlations [50,51,54]. These risks are particularly relevant in dermatology, a domain characterized by semantic overlap between conditions and diagnostic language that often encodes uncertainty [4,14]. Even light hallucinations in lesion descriptions can significantly hinder learned representations, resulting in models that underperform in downstream tasks [50,51,54]. While MM architectures and foundation models have been studied in general computer vision and other medical imaging, the effects of synthetically generated clinical notes in MM algorithms in dermatology are still not explored. This has limited the determination of conditions and mechanisms where synthetic notes can be reliably exploited [1,26,31,32].

The goal of this paper is to investigate strategies for synthesizing and leveraging clinical notes for pairing with real dermatological images, focusing on strategies that increase or reduce the negative effects of hallucinated content on learning MM dermatology data representations. The paper presents a MM architecture framework in which real images are paired with synthesized clinical notes, aiming to evaluate the following aspects: the role of metadata in prompt structure, the effect of specific keywords in the synthetic notes, the variability of different LLMs in terms of clinical note output, specific architectural designs. Each

aspect is systematically evaluated across two downstream tasks, cross-modal retrieval and zero-shot learning. The proposed MM architecture is trained on 16,000 skin lesion image-note couples (including both dermoscopic and clinical images) collected from six public data sources and evaluated on over 37,000 collected from fifteen sources that comprise the internal partitions of the six datasets used for training and nine additional external datasets.

2. Statement of significance

Problem or Issue	Dermatology datasets are often unimodal and small; using LLMs to synthesize clinical notes can create paired image–text data but risks hallucinations that introduce harmful noise into multimodal models.
What is already known	Multimodal learning improves data representation and generalization; LLMs can generate useful text but suffer domain gaps and hallucinations.
What this paper adds	A study assessing the best strategies and setups for the generation of synthetic clinical notes, investigating multiple aspects. The reliability of synthetic notes is evaluated on cross-modal retrieval and zero-shot learning.
Who would benefit from the new knowledge in this paper	Multimodal biomedical AI researchers and engineers, dataset curators, clinical AI developers/validators, and regulators focused on safe LLM use in healthcare.

3. Methods

3.1. Dataset description

The dataset is collected combining images from publicly available medical repositories. The strategy of collecting images from multiple sources aims to capture the broad visual variability of skin lesions, in terms of lighting, magnification, anatomical site, skin pigmentation, and lesion morphology, including both dermoscopic (characterized by high-resolution, centered, low-noise) and clinical (characterized by wider-field, non-magnified, possibly off-center) images. The dataset is split into training, validation, and testing partitions, as summarized in Table 1. Both dermoscopic and clinical images are included in the training and validation partitions, enabling the model to learn from heterogeneous visual inputs. The test partition includes both internal data, consisting of independent samples collected from the same sources used for training and validation, and external data collected from independent repositories, to evaluate the performance under both in-distribution and out-of-distribution conditions. All images in the dataset include skin lesions, consistent with the scope of the framework and with standard dermatology benchmarks. Skin lesions represent abnormalities in the skin by definition: a lesion, whether benign or malignant, constitutes a deviation from normal tissue and is annotated precisely because it is abnormal. The dataset therefore does not include lesion-free skin images, as such images carry no diagnostic label and serve no role in the clinical workflow. In dermatology, screening is performed on identified lesions, determining whether a lesion is benign or malignant, rather than on lesion-free skin, which falls outside the scope of any standard dermatology benchmark.

Samples used for training and validation includes dermoscopic images from BCN20000 [15] and Derm12345 [17], as well as clinical images from Derm7pt [55], DermNet [56], FLUO_SC [57], and MRA-MIDAS [58]. The test partition comprises independent datasets that were not used during training, including HAM10000 [27], SKINL2 [59], Fitzpatrick17k [60], HIBA [61] (Hospital Italiano de Buenos Aires), PAD-UFES-20 [62], SD198 [63], MSK [64], and Milk10k [65]. The same lesions included in Milk10k are acquired under both clinical and dermoscopic settings and considered as two different datasets. In addition,

Table 1

Overview of the dataset composition. The dataset includes dermatology images (paired with synthesized clinical notes) collected from multiple public available sources. Data are annotated with eight classes: BEK (benign keratosis), DF (Dermatofibroma), NEV (Benign Nevus), VASC (Vascular lesion), ACK (Pre-cancer Actinic Keratosis), BCC (Basal Cell Cancer), MEL (Melanoma), SCC (Squamous Cell Cancer). Images can be dermoscopic (D) or clinical images (C). The data are divided into three partitions: training, validation, and testing. The training and validation sets draw from six datasets, allowing the model to learn from both dermoscopic and clinical images. The testing set includes fifteen datasets in total: the test splits of those same six datasets (internal) and nine additional independent datasets (external).

Dataset	BEK	DF	NEV	VASC	ACK	BCC	MEL	SCC	Total
Training partition									
BCN20000 [15] (D)	796	86	2,944	77	515	1,966	1,999	391	8,774
Derm12345 [17] (D)	456	128	2,116	214	56	300	271	241	3,782
Derm7pt [55] (C)	46	14	18	17	0	26	160	0	281
DermNet [56] (C)	154	40	39	0	38	133	62	21	487
FLUO_SC [57] (C)	176	0	110	0	392	347	13	54	1,092
MIDAS [58] (C)	215	39	555	26	135	419	164	259	1,812
TOTAL training	1,843	307	5,782	334	1,136	3,191	2,669	966	16,228
Valid partition									
BCN20000 (D)	170	18	630	16	110	421	428	57	1,850
Derm12345 (D)	75	12	891	17	4	57	37	15	1,108
Derm7pt (C)	21	6	8	8	0	12	69	0	124
DermNet (C)	22	7	7	0	6	19	10	7	78
FLUO_SC (C)	26	0	17	0	56	50	3	9	161
MIDAS (C)	32	7	80	5	20	61	24	37	266
TOTAL validation	346	50	1633	46	196	620	571	125	3,587
Internal Testing partition									
BCN20000 (D)	172	20	632	18	112	422	430	111	1,917
Derm12345 (D)	175	40	1,733	51	14	66	92	47	2,218
Derm7pt (C)	13	4	5	5	0	7	45	0	79
DermNet (C)	44	11	11	0	11	38	18	7	140
FLUO_SC (C)	50	0	31	0	112	99	3	15	310
MIDAS (C)	61	11	158	7	38	119	47	74	515
Total Internal test	515	86	2,570	81	287	751	635	254	5,179
External Testing partition									
HAM10000 [27] (D)	1338	160	7737	180	149	622	1305	229	11,720
SKINL2 [59] (D)	33	14	97	0	0	40	28	0	212
Fitzpatrick17k [60] (C)	30	9	39	0	5	36	115	141	375
HIBA [61] (D)	88	61	602	51	63	340	253	158	1,616
PAD UFES 20 [62] (C)	235	0	244	0	730	845	52	192	2,298
SD198 [63] (C)	60	18	32	3	62	13	38	0	226
MSK [64] (D)	566	18	2595	67	251	741	1311	296	5,845
Milk10k [65] (C)	544	52	746	47	303	2522	450	473	5,137
Milk10k [65] (D)	544	52	746	47	303	2522	450	473	5,137
Total external test	3,438	384	12,838	395	1,866	7,681	4,002	1,962	32,566
TOTAL Testing	3,953	470	15,408	476	2,153	8,432	4,637	2,216	37,745

the test subset includes independent samples from the repositories used during training. From each original repository, only a subset of samples is selected. Only eight diagnostic categories are selected as classes: viz., Benign Keratosis (BEK, benign), Dermatofibroma (DF, benign), Benign Nevus (NEV, benign), Vascular Lesion (VASC, benign), Actinic Keratosis (ACK, pre-cancerous), Basal Cell Carcinoma (BCC, malignant), Melanoma (MEL, malignant), and Squamous Cell Cancer (SCC, malignant). Section S1 Dermatology-related keywords ([Supplementary Materials](#)) illustrates the mapping between the classes collected from heterogeneous datasets and the eight that are selected for the paper.

3.2. Clinical notes synthesis

The synthesis of clinical notes describing skin lesions relies on the exploitation of image-associated metadata with existing LLMs.

Publicly available dermatology datasets consist mainly of images paired with heterogeneous metadata and annotations. These metadata typically refer to lesion classes and, in some cases, finer-grained lesion descriptions focused on lesion type; however, they often lack detail about morphological characteristics. Therefore, metadata from each dataset need to be structured into a unified database including the identified lesion, the nature of the class (benign, pre-cancerous, malignant), possible lesion specialization, and additional descriptive information. Two additional fields are introduced to align lesion classes and specializations with a shared medical vocabulary, ensuring consistent textual content across sources. This vocabulary includes the eight diagnostic categories adopted in this paper and their corresponding

specializations, enabling standardized terminology and facilitating downstream multimodal learning. Aligned metadata are first exploited to generate clinical notes through a fixed templating strategy, referred to as M (metadata), where lesion classes or subclasses are used as textual descriptors to populate predefined templates. Furthermore, clinical notes are also synthesized by prompting LLMs, to generate richer textual descriptions by querying specific lesion attributes. However, LLM-generated content introduces hallucinations or additional noise, which can negatively affect MM training. Therefore, LLM usage is constrained in two ways. First, prompts are designed to emphasize both dermatologic aspects and non-specialized visual descriptors, such as asymmetry, color, and border shape. Second, aligned metadata are explicitly incorporated into the prompts to guide and condition the generation process. This metadata-conditioned strategy is referred to as L (short for LLM) and combines multiple prompts with different levels of detail to increase variability during training. Finally, LLM-based synthesis without metadata conditioning is referred to as OL (Only LLM). This configuration is included to highlight how unconstrained prompting can easily lead to hallucinated or unreliable clinical descriptions, reinforcing the importance of metadata alignment and controlled generation strategies. Four LLMs are evaluated: GPT-4o-mini, MedGemma [66], SkinGPT4 [1], and DermLIP [67]. Full prompt templates for each strategy and the fixed template structure adopted in strategy M are provided in Section S3 (Clinical note generation, [Supplementary Materials](#)).

3.3. Architectural design

The MM architecture combines visual and textual features from dermatology data within a unified framework designed to support multiple downstream tasks.

The structure explicitly models cross-modal interactions while aiming to limit overfitting, a critical issue given the limited size of the training dataset and the large number of model parameters. As illustrated in Fig. 1, the architecture is composed of two independent input

modality channels and a shared latent space. The image encoder includes PanDerm [26] encoder architecture, a large Vision Transformer trained on dermatology, receiving $224 \times 224 \times 3$ RGB images as input, processed by 24 transformer blocks. The image-encoder is fully fine-tuned during training, despite its large number of parameters, as this choice proved beneficial during architecture design. The textual branch is implemented using PubMedBERT [68], a transformer-based model trained on medical text, which processes up to 512 tokens through 12 transformer blocks. In contrast to the image encoder, the parameters of

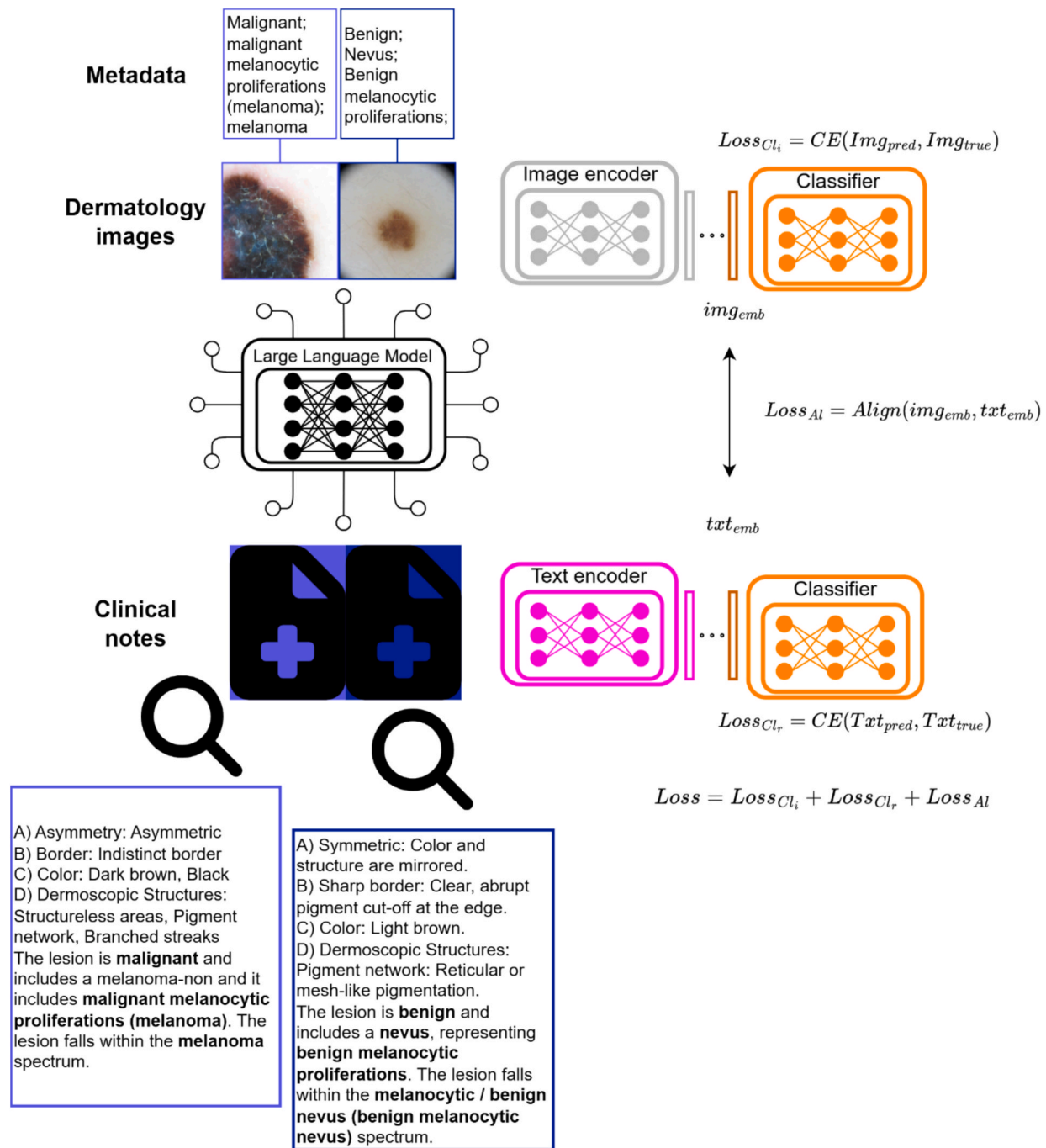


Fig. 1. Overview of the multimodal architecture, training strategy and clinical note synthesis. The architecture includes two input encoders, processing skin lesion images (grey) and clinical notes (magenta); a share projection head and classifier (orange). During the training both modalities are necessary, to align their representations. Furthermore, the training involves also the optimization of classification terms for both modalities. During the testing phase, the model can be used on the analysis of a single or multiple modalities. Clinical notes are synthesized using an LLM, using the metadata linked to original images to constrain the text generation. The Figure includes an example of generation, starting from images and metadata (shades of blue). Metadata include relevant keywords selected, which are reported in bold in the corresponding clinical notes.

the text encoder are frozen during training to ensure stable performance and reduce the risk of overfitting.

Each encoder produces a fixed-dimensional embedding that is projected into a shared latent space through an additional projection component, resulting in 128-dimensional modality-specific representations [69]. These embeddings are then fed into a shared block composed of fully connected layers, with the final layer acting as a classifier over the eight diagnostic classes. The inclusion of the classifier provides implicit regularization of the shared representation and has been shown to mitigate overfitting [70,71].

Alignment between image and text representations in the shared latent space is explicitly enforced using three complementary loss functions: an L1 loss, a cosine similarity loss, and NT-Xent [72] contrastive loss. These instance-level losses encourage embeddings from paired samples to be aligned while separating unrelated samples within the batch. When classification is included in the training setup, label information is incorporated so that samples belonging to the same class are treated as positives.

During training, embeddings from the two modalities are provided to the shared layers in an alternating manner rather than simultaneously. This strategy aims to obtain a common latent space that captures discriminative features shared across visual and textual inputs. Despite the challenges posed by dataset size and model complexity, the learned shared representation enables tasks that are not directly optimized during training, such as cross-modal retrieval, zero-shot learning, and clinical note generation.

3.4. Experiment description

Synthetic data produced by LLM are inherently noisy and include hallucinations. The scope of the paper is not to qualitatively assess hallucinated notes, but to evaluate how they hinder the training of a MM architecture when paired with real dermatology images. The present analysis of hallucinations in synthesized clinical notes is conducted at the label level, evaluating whether the generated note includes a reference to the correct diagnostic class (e.g., melanoma, basal cell carcinoma). Section S6 (Supplementary Materials) describes the number of correct generated notes, considering the lesion class. A broader assessment of hallucinations may extend to additional aspects of clinical content, including dermoscopic descriptors, lesion attributes, anatomical localization, and patient history elements, and would involve systematic clinical review of the generated notes, which is not feasible considering our corpus (including hundreds of thousands of clinical notes).

3.4.1. Experiments on the role of metadata to design LLM prompts

This set of experiments aims to evaluate the reliability of synthesized clinical notes, and their impact on multimodal alignment, under the hypothesis that they reduce hallucinations and limit the propagation of noisy content in clinical notes.

This set of experiments involves four strategies to synthesize clinical notes: **M** (metadata used to fill a fixed note template), **OL** (clinical notes synthesized with a LLM, without any metadata into the prompt), **L** (clinical notes synthesized with a LLM, introducing metadata into the prompt), **LM** (clinical notes generated with L and M strategies). M strategy is used as a baseline, while OL is adopted to highlight how the absence of metadata conditioning can lead to unreliable and heterogeneous textual descriptions. L strategy is adopted to show that introducing metadata in the LLM prompt leads to less noisy synthetic clinical notes. Finally, the LM strategy evaluates whether the combination of complementary clinical notes improves robustness compared to individual strategies. The strategies are evaluated on the synthetic notes generated by SkinGPT4, pre-trained on dermatology data.

3.4.2. Experiments on the effect of specific keywords in clinical notes

This set of experiments aims to evaluate how keywords related to

dermatology aspects influence the training, providing two setups where the keyword presence is either amplified or reduced. Both cases involve the MM architecture trained with strategy LM, considering clinical notes synthesized by SkinGPT4. In the first case (defined as KEY), the presence of specific keyword is amplified adding an additional loss function during the training, which aims to align the image representation to the representation of one or more keywords, together with the alignment between the same image and the clinical note. The set of keywords is shown in Section S1. Dermatology-related keywords (Supplementary Materials). The underlying hypothesis is that aligning images not only with clinical notes but also with relevant keywords enables the model to learn more detailed and discriminative representations. On the other hand, noisy keywords lead the model to align images to the wrong dermatology concept, and therefore to a poorer MM representation. The second setup (defined as FILT) aims to cover this aspect, removing subsentences from the clinical notes, limiting the propagation of wrong content regarding the lesion to the image representation. The description of keyword selection and sentence filtering is explored in S5 Keyword extraction and sentence filtering strategies (Supplementary Materials).

3.4.3. Experiments on the variability of different LLMs

This set of experiments aims to evaluate how clinical notes generated by different LLMs lead to different performance. The experiments involve the comparison of four LLMs and their combination (i.e. defined as whole), trained on different data: Gpt-4o-mini is trained on general purpose data, therefore it is not specialized on medical or dermatology data; MedGemma is trained on biomedical data (images and texts); SkinGPT4 and DermLIP are trained on dermatology data (images and texts). The hypothesis is that LLMs specialized on dermatology lead to less robust clinical notes, including less hallucinations. However, the corpora on which LLMs are trained is not the only relevant factor. The LLM architectural design influences the output as well. Gpt and MedGemma, in particular, can be prompted with queries including more tokens (i.e. more words). They can output longer sequences and are trained on a larger variety of prompts. In contrast, SkinGPT4 and DermLIP are prompted with a small subset of queries and provide limited output sequences.

3.4.4. Experiments on architectural design

This set of experiments aims to analyze which training strategy better aligns image and text representations without involving classifier optimization. The architectural design including a classifier aims to reduce the overfitting and leads to higher performance. However, current trend in foundation models is based on unsupervised training that aims to collect larger datasets without the need of annotations. The architecture is trained to align images and synthetic notes using strategy LM, using an LLM to generate them.

3.5. Downstream tasks

The downstream tasks considered in this paper, cross-modal retrieval and zero-shot learning, are inherently multimodal and are not designed to quantify performance improvement over unimodal image-based classifiers. Cross-modal retrieval requires shared image-text representation by design. Zero-shot learning, despite involving a classification decision, does not rely on label supervision during training, generalizing instead through alignment between image and textual concept embeddings. The experiments evaluate the conditions under which synthetic notes produce reliable multimodal alignment, rather than the absolute performance gain over conventional supervised approaches, which would require a different experimental design and task framework.

3.5.1. Cross-modal retrieval

The cross-modal retrieval task assesses the system capability to identify relevant samples in one modality based on a query from the

other modality. Embeddings are extracted from the shared latent space (immediately preceding the classifier), and cosine similarity is used to rank candidates, where higher similarity indicates increased relevance. Retrieval performance is measured in both directions: Image-to-Text (retrieving reports for a query image) and Text-to-Image (retrieving images given a query report). In the Image-to-Text setting, the pool of candidate reports includes clinical notes synthesized with M strategy. For Text-to-Image retrieval, the pool includes images. Performance is computed using micro-average precision (mAP), which aggregates retrieval results across all queries into a single prediction set.

3.5.2. Zero-shot learning

Zero-shot learning assesses the system capability to link images with classes or concepts that are not explicitly observed by the classifier during supervised training. Here as well, the image embeddings from the shared latent space are compared against embeddings of candidate concepts using cosine similarity, and the concept with the highest similarity is assigned. Two evaluation setups are considered: (i) a multiclass setting using the five diagnostic categories, referred herein as “8-classes” setup; and (ii) an extended semantic setting that includes a broader list of dermatology-related terms: ‘actinic keratosis’, ‘basal cell cancer’, ‘benign/seborrheic keratosis’, ‘dermatofibroma’, ‘melanocytic / benign nevus’, ‘melanoma’, ‘squamous cell cancer’, ‘vascular lesion’. Regarding the subclass specialization, the medical concepts that are adopted for the mapping are: ‘actinic keratosis’, ‘basal cell carcinoma’, ‘benign melanocytic nevus’, ‘blue nevus’, ‘bowen disease / SCC in situ’, ‘congenital / special-pattern nevi’, ‘dermatofibroma / fibrous lesions’, ‘dysplastic / atypical nevus (clark-type)’, ‘lentigo maligna’, ‘lichenoid keratosis’, ‘melanoma’, ‘seborrheic keratosis & pigmented keratoses solar lentigo’, ‘squamous cell carcinoma (invasive)’, ‘vascular lesions’), referred herein as “15-classes” setup. This setting includes finer-grained semantic associations that were not included during training as classes. Results are reported in terms of weighted F1-score.

3.6. Data pre-processing and augmentation

Data pre-processing is performed for both visual and textual data, to produce inputs compatible with the respective encoders. For the image branch, regions including lesions are automatically cropped and resized to the 224×224 input expected by the visual backbone. The pre-processing pipeline differs by acquisition type: dermoscopic images undergo segmentation to produce a lesion mask using Otsu thresholding, to identify the region to crop; clinical (non-dermoscopic) images are manually extracted using manually annotated bounding boxes. For the textual branch, notes are processed into BERT tokens, with sequences truncated to the BERT maximum of 512 elements. Data augmentation is applied to the images during training, to increase model robustness. Transformations included geometric operations (90° rotations and horizontal and vertical flips) together with color perturbations applied to the RGB channels (perturbation range -50 to 10). To mitigate class imbalance and improve generalization, we employed class-wise data augmentation (CWDA), oversampling under-represented classes by presenting augmented variants of those samples more frequently during training. No augmentation was applied to clinical notes.

3.7. Experimental setup

The experimental setup jointly addresses hyperparameter selection. Hyperparameters are selected using a grid search and fixed for all reported experiments: models are trained for 15 epochs with a learning rate of $1e-4$, a decay rate of $1e-5$, batch size of 32. While all the training loss functions are not weighted, the alignment loss functions are weighted to 0.5 (NT-Xent) and 1 (L1- and Cosine Similarity Loss); the temperature parameters are set to 0.5 and 0.07, respectively.

3.8. Hardware and Software

Deep learning algorithms are developed and trained with PyTorch 2.60, and all experiments are implemented using Python libraries. The BERT architecture is implemented using Transformers 4.52.4, which is also employed to pre-process textual reports. Image pre-processing and augmentation are carried out with albumentations 2.0.5. Numpy 2.0.2 is used to support other operations during training. Model performance is evaluated using the metrics provided by scikit-learn 0.23. All deep learning experiments are conducted on a V100x GPU. Clinical notes are synthesized using multiple libraries, in particular open 1.74.0 to use GPT-4o-mini as backbone model (temperature set to 0.2, 250 tokens as output). The other LLMs are implemented considering the following repositories: MedGemma (<https://huggingface.co/google/medgemma-4b-it>), SkinGPT4 (<https://github.com/JoshuaChou2018/SkinGPT-4?tab=readme-ov-file>), DermLIP (<https://huggingface.co/moxeeem/dermlip-gpt2-captioner>).

4. Results

The Results section assess the model performance considering the experiments described in Section 2.4 Experiment description, aggregating the performance on five sets, considering the testing partition: Internal, the datasets belonging to the internal partition; External, the datasets belonging to the external partition; Dermoscopic, the datasets containing dermoscopic images (marked with D in Table 1); Clinic, the datasets including clinic images (marked with C in Table 1); Whole, including the whole testing partition. Fig. 2 summarizes the key findings, indicating that the LM strategy is the most effective approach for synthesizing notes (Section 3.1), and that the combination (Whole) of synthetic notes generated by the four LLMs (Section 3.3) provides the optimal solution. A qualitative analysis of misclassification and retrieval error patterns across strategies is provided in Section S7 (Qualitative Results, Supplementary Materials).

4.1. The role of metadata to design LLM prompts

Introducing metadata into LLM prompts consistently reduces noisy or hallucinated content and improves multimodal alignment. Table 2 reports the performance obtained on cross-modal retrieval and zero-shot learning tasks when training the MM architecture with clinical notes synthesized using different strategies. Results consistently show that strategies including metadata in the prompt (L and LM) reach the highest performance, in both downstream tasks, with LM strategy leading to the highest performance. This suggests that the MM architecture benefits from note variability obtained combining notes generated with M and L strategies. While M achieves high performance in the retrieval tasks, mostly because the notes used for evaluation are the ones generated with M strategy, the performance on zero-shot learning is poor. Clinical notes generated without metadata conditioning (OL) systematically underperform all other strategies in both cross-modal retrieval and zero-shot learning, confirming that unconstrained LLM generation introduces noisy and unreliable textual descriptions that hinder learning.

The M strategy, which directly inserts lesion class or subclass into a fixed template, represents the most explicit case of label verbalization and therefore serves as a natural upper bound for potential label leakage. Despite this, M does not achieve the highest performance. It is consistently outperformed by LM across both retrieval and zero-shot tasks, and by L in the zero-shot setting. In particular, the results on cross-modal retrieval are relevant, since they are evaluated on the clinical notes synthesized with M. This suggests that performance improvement cannot be attributed to label verbalization alone. Furthermore the zero-shot task, where image embeddings are matched against textual concept embeddings not seen as training classes, would be expected to reach values close to ceiling, while instead, even the

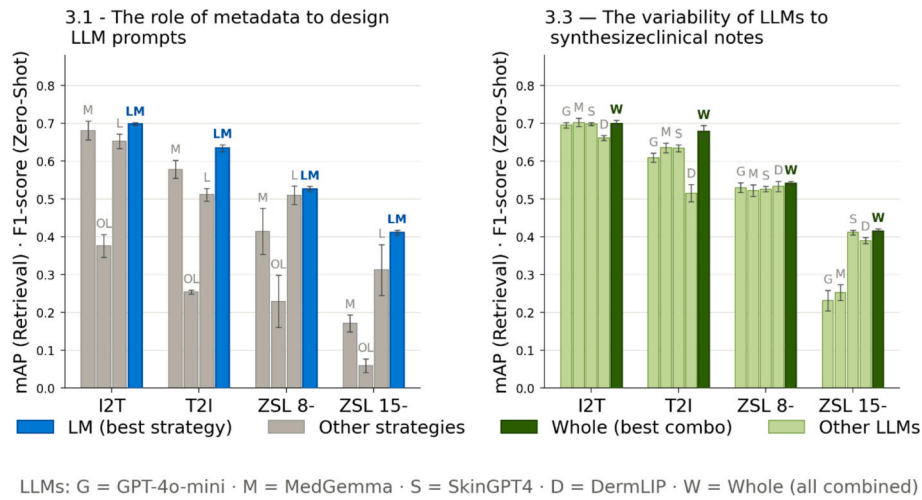


Fig. 2. Overview of the main results for selecting the optimal strategy to synthesize clinical notes (Section 3.1, left) and the LLM to use for synthesis (Section 3.3, right). In both settings, performance is reported on cross-modal retrieval (Image-to-Text, I2T; Text-to-Image, T2I) and zero-shot learning (ZSL; 8 classes and 15 classes).

Table 2

Overview of the results evaluated on cross-modal retrieval (both image-to-text and text-to-image setups) and zero-shot learning (both 8- and 15-classes setups). Both tasks are evaluated considering the performance of the MM architecture, trained with clinical notes generated with multiple strategies: M, clinical notes generated filling a note template with metadata; OL, clinical notes synthesized with a LLM, without any metadata used in the prompt; L, clinical notes synthesized with a LLM, providing metadata in the prompt; LM, clinical notes generated with L and M. Performance is evaluated on fifteen datasets (six internal, nine external), reporting the mAP for cross-modal retrieval task, weighted F1-score for the zero-shot learning task. Datasets are aggregated in five partitions: internal (datasets from internal testing partition), external (datasets from external testing partition), Dermoscopic (testing partition datasets including dermoscopic images), Clinic (testing partition datasets including clinic images), Whole (whole testing partition).

Partition	M	OL	L	LM	M	OL	L	LM
Cross-modal retrieval: Image-to-Text					Cross-modal retrieval: Text-to-Image			
Internal	0.791 ± 0.039	0.444 ± 0.039	0.759 ± 0.019	0.806 ± 0.004	0.766 ± 0.029	0.282 ± 0.013	0.636 ± 0.019	0.778 ± 0.003
External	0.663 ± 0.023	0.365 ± 0.028	0.635 ± 0.019	0.681 ± 0.005	0.548 ± 0.023	0.250 ± 0.005	0.491 ± 0.019	0.611 ± 0.010
Dermoscopic	0.718 ± 0.022	0.419 ± 0.036	0.692 ± 0.017	0.732 ± 0.006	0.555 ± 0.020	0.262 ± 0.005	0.521 ± 0.023	0.619 ± 0.011
Clinic	0.592 ± 0.029	0.302 ± 0.017	0.562 ± 0.022	0.612 ± 0.008	0.689 ± 0.033	0.249 ± 0.006	0.526 ± 0.036	0.712 ± 0.007
Whole	0.681 ± 0.025	0.376 ± 0.030	0.652 ± 0.019	0.698 ± 0.004	0.578 ± 0.024	0.254 ± 0.005	0.511 ± 0.017	0.634 ± 0.009
Zero-shot: 8-classes					Zero-shot: 15-classes			
Internal	0.409 ± 0.101	0.229 ± 0.097	0.554 ± 0.021	0.564 ± 0.013	0.258 ± 0.025	0.075 ± 0.012	0.251 ± 0.041	0.315 ± 0.009
External	0.415 ± 0.055	0.228 ± 0.065	0.503 ± 0.024	0.520 ± 0.007	0.165 ± 0.023	0.057 ± 0.020	0.322 ± 0.072	0.429 ± 0.007
Dermoscopic	0.448 ± 0.076	0.261 ± 0.092	0.559 ± 0.023	0.576 ± 0.007	0.151 ± 0.014	0.060 ± 0.020	0.316 ± 0.089	0.434 ± 0.006
Clinic	0.366 ± 0.024	0.195 ± 0.032	0.434 ± 0.028	0.451 ± 0.009	0.224 ± 0.036	0.062 ± 0.019	0.268 ± 0.022	0.288 ± 0.008
Whole	0.414 ± 0.061	0.229 ± 0.069	0.510 ± 0.024	0.526 ± 0.007	0.171 ± 0.022	0.059 ± 0.018	0.312 ± 0.067	0.411 ± 0.006

strongest strategy (LM) achieves a weighted F1 of 0.526 and 0.411 on the 8- and 15-class setups, well below what pure label propagation would produce.

4.2. The effects of specific keywords

Including dermatology-related keywords in synthetic notes has only a modest effect on downstream MM training.

Table 3 compares the LM baseline with two variants: KEY, which amplifies keyword influence via an auxiliary alignment loss, and FILT, which reduces keyword presence by filtering subsentences. Across cross-modal retrieval and zero-shot evaluations the three variants produce largely similar metrics, with no clear, consistent advantage for keyword amplification. Small variations appear across partitions but lack a systematic pattern, suggesting the model is relatively robust to explicit keyword supervision. A possible explanation is that the image-text alignment objective and the MM encoder implicitly learn to filter irrelevant or noisy text tokens, preventing noisy keywords from dominating the learned representation. FILT shows that removing suspicious textual fragments does not markedly harm performance either, indicating the architecture can tolerate moderate levels of textual noise.

The robustness of MM performance to keyword amplification (KEY)

and keyword reduction (FILT) further contributes to validating the findings involving label leakage. Explicitly amplifying diagnostic keywords does not lead to consistent improvements across tasks and partitions, while reducing their presence does not lead to consistent degradation. The learned representations are therefore not driven by keyword presence as a shortcut, suggesting that the model aligns image-text representations, rather than learning only to encode labels in the text branch.

4.3. The variability of LLMs to synthesize clinical notes

Clinical notes synthesized by different LLMs yield broadly comparable MM performance.

Table 4 reports cross-modal retrieval and zero-shot metrics for models trained using notes from gpt-4o-mini (general), MedGemma (biomedical), SkinGPT4 and DermLIP (dermatology-specialized), and their combination (Whole). Overall, differences across LLMs are small: image-to-text retrieval mAP and most zero-shot scores are similar, indicating that the MM training pipeline is robust to variations in textual style and length. Two observations emerge from the results. First, combining notes from multiple LLMs (Whole) often matches or slightly improves average performance, suggesting complementary strengths

Table 3

Overview of the results evaluated on cross-modal retrieval (both image-to-text and text-to-image setups) and zero-shot learning (both 8- and 15-classes setups). Both tasks are evaluated considering the performance of the MM architecture, trained with clinical notes generated with LM strategy. The set of experiments analyzes the effect of dermatology-related keywords in the synthesized clinical notes, considering setups in which their influence is either amplified (KEY) or reduced through filtering (FILT). Performance is evaluated on fifteen datasets (six internal, nine external), reporting the mAP for cross-modal retrieval task, weighted F1-score for the zero-shot learning task. Datasets are aggregated in five partitions: internal (datasets from internal testing partition), external (datasets from external testing partition), Dermoscopic (testing partition datasets including dermoscopic images), Clinic (testing partition datasets including clinic images), Whole (whole testing partition).

Partition	LM	KEY	FILT	LM	KEY	FILT
	Cross-modal retrieval: Image-to-Text			Cross-modal retrieval: Text-to-Image		
Internal	0.806 ± 0.004	0.790 ± 0.008	0.796 ± 0.007	0.778 ± 0.003	0.772 ± 0.008	0.758 ± 0.001
External	0.681 ± 0.005	0.680 ± 0.005	0.684 ± 0.007	0.611 ± 0.010	0.608 ± 0.027	0.671 ± 0.017
Dermoscopic	0.732 ± 0.006	0.730 ± 0.004	0.737 ± 0.007	0.619 ± 0.011	0.615 ± 0.028	0.690 ± 0.018
Clinic	0.612 ± 0.008	0.612 ± 0.009	0.613 ± 0.007	0.712 ± 0.007	0.712 ± 0.011	0.692 ± 0.009
Whole	0.698 ± 0.004	0.695 ± 0.006	0.700 ± 0.007	0.634 ± 0.009	0.631 ± 0.024	0.683 ± 0.015
	Zero-shot: 8-classes			Zero-shot: 15-classes		
Internal	0.564 ± 0.013	0.553 ± 0.010	0.560 ± 0.004	0.315 ± 0.009	0.280 ± 0.021	0.314 ± 0.007
External	0.520 ± 0.007	0.512 ± 0.012	0.519 ± 0.006	0.429 ± 0.007	0.404 ± 0.021	0.428 ± 0.007
Dermoscopic	0.576 ± 0.007	0.570 ± 0.012	0.570 ± 0.006	0.434 ± 0.006	0.420 ± 0.018	0.433 ± 0.005
Clinic	0.451 ± 0.009	0.436 ± 0.014	0.454 ± 0.005	0.288 ± 0.008	0.244 ± 0.030	0.291 ± 0.008
Whole	0.526 ± 0.007	0.553 ± 0.010	0.524 ± 0.005	0.411 ± 0.006	0.386 ± 0.021	0.412 ± 0.006

across generators. Second, dermatology-specialized LLMs (SkinGPT4, DermLIP) tend to perform as well or better than general models in certain zero-shot settings (SkinGPT4 notably excels on the 15-class zero-shot task) implying that domain-specific pretraining can reduce clinically relevant hallucinations despite shorter prompts/outputs.

4.4. The architectural design

Optimizing a classification head during training substantially improves downstream performance compared with purely unsupervised alignment.

Table 5 compares the performance obtained on cross-modal retrieval and zero-shot learning tasks when training the MM architecture with three variants: LM with classifier optimization (LM (CI)), and both L and LM trained without optimizing the classifier (No CI). The classifier-optimized model reaches significantly higher mAP and zero-shot F1 across all partitions. This indicates that exploiting annotations, when available, stabilizes MM learning, reducing overfitting to noisy text, and increasing performance on downstream tasks. Furthermore, LM trained without a classifier still outperforms L, suggesting a clear development path: while classifier optimization yields the best immediate performance, the LM strategy provides a viable, label-efficient foundation for scaling to larger, annotation-sparse datasets.

5. Discussion

This work demonstrates that synthetic clinical notes can be effectively leveraged to train reliable multimodal MM architectures and foundation models for dermatology. The analysis focuses on several aspects that may influence the reliability of multimodal alignment, including the adoption of metadata, the effects of dermatology-related keywords, the variability of notes synthesized by different LLMs, and the importance of specific architectural designs.

The main advantage of synthetic clinical notes involves their ability to overcome the limitations of real clinical documentation, which is costly to produce, difficult to standardize, and often inaccessible due to privacy constraints. By adopting large-scale pairing of images with textual descriptions, synthetic notes remove the need for the involvement of dermatologists in the annotation loop, which is time-consuming. At the same time, synthetic notes are inherently noisy and include hallucinated or inaccurate content, which can hinder the learning of MM architectures by introducing spurious cross-modal associations. The hallucination analysis presented in this work is necessarily focused on label-level content, specifically whether a generated note contains a reference to the correct diagnostic class, as this is the only aspect of clinical accuracy that can be verified automatically, without requiring

expert clinical annotation of the generated corpus. A direct comparison of the semantic distribution of synthetic and real clinical note embeddings was not feasible due to the absence of suitable ground-truth text in the domain. To mitigate overfitting to repetitive synthetic patterns, note diversity was deliberately increased by combining fixed-template and LLM-generated notes across multiple models with distinct pretraining corpora, a strategy whose effectiveness is reflected in the downstream performance difference reported in the Results. Given the scale of the generated corpus, including over 55,000 images across four LLMs and multiple prompting strategies, direct qualitative assessment of hallucinations is infeasible; their impact is therefore evaluated indirectly through performance on cross-modal retrieval and zero-shot learning.

Results show that synthetic notes can support effective MM training when generated under specific conditions. Clinical notes that are generated by prompting the LLMs with queries including image metadata (L and LM) consistently lead to stronger performance. However, LLMs introduce variability and allow querying specific lesion attributes, enriching textual descriptions, while metadata ensure that the lesion identity and its clinical nature are preserved. This grounding effect propagates from the lesion type to related lesion characteristics, limiting clinically relevant hallucinations. The robustness of this setup is further supported by experiments on dermatology-related keywords, where neither explicit amplification nor filtering significantly alters performance, indicating that the model can mitigate residual textual noise during alignment. The choice of LLM also influences the performance. Dermatology-specific LLMs, despite being built on an architectural design supporting shorter prompts and outputs, achieve more robust results than general-purpose models, suggesting that domain-specific pretraining leads to more fine-grained clinical features that cannot be fully compensated by verbose prompting alone. Importantly, high performance on external datasets indicates that synthetic text also acts as a form of regularization. Exposure to heterogeneous descriptions encourages the model to learn more abstract, semantically grounded representations, reducing overfitting to dataset-specific visual patterns and improving out-of-distribution generalization. This is further supported by the analysis of note generation strategies and keyword supervision, which consistently indicates that the quality and clinical relevance of the textual descriptions, rather than the mere presence of diagnostic labels or keywords, drives multimodal alignment. The M strategy, which represents the most direct form of label verbalization, does not achieve the highest performance, and explicitly amplifying diagnostic keywords produces no consistent gain, while reducing their presence produces no consistent degradation. These findings suggest that the MM architecture exploits the semantic richness introduced by metadata-conditioned notes, rather than relying on label encoding in the text branch as a shortcut to downstream performance.

Table 4

Overview of the results evaluated on cross-modal retrieval (both image-to-text and text-to-image setups) and zero-shot learning (both 8- and 15-classes setups). Both tasks are evaluated considering the performance of the MM architecture, trained with clinical notes generated with multiple LM strategy, considering four LLMs and their combination: gpt-4o-mini, MedGemma, SKinGPT4, DermLIP. The model trained using all the clinical notes during the training is referred to as Whole. Performance is evaluated on fifteen datasets (six internal, nine external), reporting the mAP for cross-modal retrieval task, weighted F1-score for the zero-shot learning task. Datasets are aggregated in five partitions: internal (datasets from internal testing partition), external (datasets from external testing partition), Dermoscopic (testing partition datasets including dermoscopic images), Clinic (testing partition datasets including clinic images), Whole (whole testing partition).

Partition	GPT	MedGemma	SkinGPT4	DermLIP	Whole	GPT	MedGemma	SkinGPT4	DermLIP	Whole
	Cross-modal retrieval: Image-to-Text					Cross-modal retrieval: Text-to-Image				
Internal	0.797 ± 0.011	0.807 ± 0.006	0.806 ± 0.004	0.756 ± 0.016	0.792 ± 0.007	0.775 ± 0.007	0.783 ± 0.009	0.778 ± 0.003	0.588 ± 0.019	0.758 ± 0.01
	0.679 ± 0.007	0.685 ± 0.012	0.681 ± 0.005	0.646 ± 0.007	0.684 ± 0.009	0.582 ± 0.013	0.612 ± 0.014	0.611 ± 0.010	0.503 ± 0.024	0.665 ± 0.018
Dermoscopic	0.731 ± 0.004	0.735 ± 0.012	0.732 ± 0.006	0.700 ± 0.007	0.732 ± 0.008	0.589 ± 0.013	0.620 ± 0.014	0.619 ± 0.011	0.558 ± 0.017	0.680 ± 0.020
	0.607 ± 0.012	0.620 ± 0.010	0.612 ± 0.008	0.574 ± 0.013	0.614 ± 0.012	0.705 ± 0.011	0.716 ± 0.009	0.712 ± 0.007	0.449 ± 0.041	0.701 ± 0.012
Whole	0.695 ± 0.007	0.702 ± 0.011	0.698 ± 0.004	0.661 ± 0.007	0.699 ± 0.009	0.609 ± 0.012	0.635 ± 0.013	0.634 ± 0.009	0.515 ± 0.023	0.678 ± 0.016
	Zero-shot: 8-classes					Zero-shot: 15-classes				
Internal	0.557 ± 0.019	0.552 ± 0.021	0.564 ± 0.013	0.565 ± 0.020	0.568 ± 0.007	0.301 ± 0.028	0.317 ± 0.018	0.315 ± 0.009	0.287 ± 0.023	0.29 ± 0.013
	0.526 ± 0.012	0.518 ± 0.014	0.520 ± 0.007	0.527 ± 0.014	0.537 ± 0.007	0.227 ± 0.027	0.249 ± 0.022	0.429 ± 0.007	0.408 ± 0.006	0.435 ± 0.007
Dermoscopic	0.578 ± 0.015	0.567 ± 0.018	0.576 ± 0.007	0.579 ± 0.016	0.587 ± 0.006	0.209 ± 0.029	0.231 ± 0.027	0.434 ± 0.006	0.406 ± 0.010	0.437 ± 0.007
	0.453 ± 0.014	0.449 ± 0.017	0.451 ± 0.009	0.465 ± 0.014	0.466 ± 0.010	0.291 ± 0.024	0.298 ± 0.018	0.288 ± 0.008	0.277 ± 0.013	0.285 ± 0.011
Whole	0.53 ± 0.013	0.522 ± 0.015	0.526 ± 0.007	0.533 ± 0.014	0.541 ± 0.006	0.231 ± 0.027	0.252 ± 0.021	0.411 ± 0.006	0.390 ± 0.008	0.414 ± 0.007

These findings have both clinical and research implications. Clinically, MM architectures can support telemedicine and documentation workflows, for example assisting dermatologists in drafting reports or enabling text-based querying of image archives in remote or resource-limited settings. These capabilities are particularly relevant in contexts where specialist availability is limited. From a research perspective, the results provide a foundation for scaling multimodal dermatology models beyond the constraints of manually image-text paired datasets. However, translating these capabilities into reliable clinical tools requires acknowledging practical and methodological limitations that limit the scope of the present work. The hallucination analysis is focused on label-level accuracy and does not account for other forms of clinically hallucinated content, including inaccurate dermoscopic descriptors, lesion attributes, anatomical localization, or patient history elements. A comprehensive assessment would require systematic clinical review of the generated notes, which is beyond the scope of this work. The evidence presented therefore supports the claim that metadata conditioning reduces label-level hallucinations and improves downstream

multimodal performance but does not cover the presence of other forms of hallucinated content in the synthesized notes. A related practical consideration involves the computational overhead introduced by synthetic note generation. Querying a LLM for each image in a large dataset is a time-consuming process and raises privacy-related concerns when proprietary models are adopted, as image metadata must be transmitted to external services. Medical and dermatology-specific LLMs are publicly available and can be deployed locally without significant computational resources, mitigating both concerns. GPT-4o-mini, while proprietary, introduces an additional monetary cost; however, at approximately \$0.40 per million input tokens and \$1.60 per million output tokens, the cost of synthesizing notes for one thousand images amounts to approximately \$2, which remains modest relative to the cost of manual clinical annotation. Importantly, note synthesis represents a one-time offline process: once generated, the notes can be reused across experiments without additional overhead. From a research perspective, future work should further refine architectural design and prompting strategies. Although classifier optimization remains important for strong

Table 5

Overview of the results evaluated on cross-modal retrieval (both image-to-text and text-to-image setups) and zero-shot learning (both 8- and 15-classes setups). Both tasks are evaluated considering the performance of the MM architecture, trained with clinical notes generated by SkinGPT4. Three training variants are compared: LM with a classifier (CI), L and LM trained without optimizing the classifier. Performance is evaluated on fifteen datasets (six internal, nine external), reporting the mAP for cross-modal retrieval task, weighted F1-score for the zero-shot learning task. Datasets are aggregated in five partitions: internal (datasets from internal testing partition), external (datasets from external testing partition), Dermoscopic (testing partition datasets including dermoscopic images), Clinic (testing partition datasets including clinic images), Whole (whole testing partition).

Partition	LM (CI)	L (No CI)	LM (No CL)	LM (CI)	L (No CI)	LM (No CL)
	Cross-modal retrieval: Image-to-Text			Cross-modal retrieval: Text-to-Image		
Internal	0.806 ± 0.004	0.260 ± 0.012	0.616 ± 0.016	0.778 ± 0.003	0.313 ± 0.037	0.630 ± 0.034
External	0.681 ± 0.005	0.238 ± 0.008	0.555 ± 0.022	0.611 ± 0.010	0.258 ± 0.023	0.462 ± 0.029
Dermoscopic	0.732 ± 0.006	0.258 ± 0.011	0.593 ± 0.022	0.619 ± 0.011	0.293 ± 0.032	0.479 ± 0.027
Clinic	0.612 ± 0.008	0.219 ± 0.005	0.490 ± 0.022	0.712 ± 0.007	0.223 ± 0.014	0.556 ± 0.039
Whole	0.698 ± 0.004	0.241 ± 0.008	0.564 ± 0.021	0.634 ± 0.009	0.265 ± 0.024	0.485 ± 0.029
	Zero-shot: 8-classes			Zero-shot: 15-classes		
Internal	0.564 ± 0.013	0.017 ± 0.008	0.405 ± 0.055	0.315 ± 0.009	0.014 ± 0.005	0.137 ± 0.040
External	0.520 ± 0.007	0.020 ± 0.001	0.402 ± 0.040	0.429 ± 0.007	0.013 ± 0.011	0.181 ± 0.062
Dermoscopic	0.576 ± 0.007	0.019 ± 0.008	0.458 ± 0.050	0.434 ± 0.006	0.013 ± 0.014	0.194 ± 0.078
Clinic	0.451 ± 0.009	0.021 ± 0.006	0.291 ± 0.043	0.288 ± 0.008	0.011 ± 0.081	0.095 ± 0.052
Whole	0.526 ± 0.007	0.020 ± 0.010	0.402 ± 0.042	0.411 ± 0.006	0.013 ± 0.094	0.175 ± 0.059

performance, scaling to larger datasets may enable fully label-free training through modality alignment alone. In parallel, post-filtering or iterative refinement of synthetic notes represents a promising direction to reinforce a virtuous cycle of synthetic data generation and MM model improvement. Addressing the remaining hallucination dimensions through structured clinical annotation protocols or automated fact-checking pipelines grounded in dermatological knowledge bases represents an important direction for future work.

6. Conclusion

This paper shows that synthesized clinical notes can be reliably paired with real dermatological images to train MM architectures and foundation models. Synthetic notes allow multimodal learning to scale beyond the limitations of real clinical documentation. However, they are inherently noisy and include hallucinated content. The paper demonstrates under which conditions synthesized clinical notes can be adopted: notes generated prompting LLMs with metadata emerge as a key factor to reduce hallucinated content and stabilize MM alignment, enabling robust performance across cross-modal retrieval and zero-shot learning tasks. Under these conditions, synthetic clinical text represents a valuable resource for advancing MM dermatology models, supporting both future research on label-efficient learning and the development of assistive clinical tools.

CRedit authorship contribution statement

Niccolo Marini: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhaohui Liang:** Writing – review & editing, Validation. **Sivaramakrishnan Rajaraman:** Writing – review & editing, Validation. **Zhiyun Xue:** Writing – review & editing, Validation. **Sameer Antani:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The code to implement the multimodal architecture and to synthesize the clinical notes are publicly available: <https://github.com/antani-lab/Synthesized-Clinical-Notes-Multimodal-AI-Models>.

All datasets are publicly available:

- BCN20000: <https://api.isic-archive.com/collections/249/>
- derm12345: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DAXZ7P>
- Derm7pt: <https://derm.cs.sfu.ca/Welcome.html>
- DermNet: <https://www.kaggle.com/datasets/shubhamgoel27/dermnet>
- FLUO_SC: <https://data.mendeley.com/datasets/s8n68jj678/1>

- MRA_MIDAS: <https://aimi.stanford.edu/datasets/mra-midas-Multimodal-Image-Dataset-for-AI-based-Skin-Cancer>
- HAM10000: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T>
- SKINL2: <https://www.it.pt/AutomaticPage?id=3459>
- Fitzpatrick17k: <https://github.com/mattgroh/fitzpatrick17k>
- HIBA: <https://api.isic-archive.com/collections/251/>
- PAD UFES 20: <https://data.mendeley.com/datasets/zr7vgbcy2/1>
- SD198: <https://huggingface.co/datasets/resyhgerwshshgdfghsdfgh/SD-198>
- MSK: composed from <https://api.isic-archive.com/collections/289/>, <https://api.isic-archive.com/collections/290/>, <https://api.isic-archive.com/collections/288/>, <https://api.isic-archive.com/collections/287/>, <https://api.isic-archive.com/collections/286/>
- Milk10k: <https://challenge.isic-archive.com/landing/milk10k/>

Partition splitting and pre-processing methods are included in the repository.

Ethics statement

- The work described has not been published previously.
- The article is not under consideration for publication elsewhere.
- The article's publication is approved by all authors.
- If accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright-holder.
- Datasets were collected from public repositories.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors used gpt-4.1-mini to generate clinical notes from input images. This relates to the main content of the research described in the manuscript. During the preparation of the manuscript, Gpt-5 was used to improve the text. Afterwards, the authors reviewed and edited the LLM. The authors take full responsibility for the content of the published article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbi.2026.105055>.

Data availability

Publicly available data sets were used. Links provided in manuscript.

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